IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

COMMUNICATIONS

Monday, 27 April 2026, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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EXAM QUESTIONS

Information for Students

Fourier Transform Pairs

Pair	$x(t)$	$X(f)$
1.	$\Pi\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc} \tau f$
2.	$2W \operatorname{sinc} 2Wt$	$\Pi\left(\frac{f}{2W}\right)$
3.	$\Lambda\left(\frac{t}{\tau}\right)$	$\tau \operatorname{sinc}^2 \tau f$
4.	$e^{-\alpha t}\, u(t),\ \alpha > 0$	$\frac{1}{\alpha + j2\pi f}$
5.	$t\, e^{-\alpha t}\, u(t),\ \alpha > 0$	$\frac{1}{(\alpha + j2\pi f)^2}$
6.	$e^{-\alpha \lvert t \rvert},\ \alpha > 0$	$\frac{2\alpha}{\alpha^2 + (2\pi f)^2}$
7.	$e^{-\pi (t/\tau)^2}$	$\tau\, e^{-\pi (f\tau)^2}$
8.	$\delta(t)$	1
9.	1	$\delta(f)$
10.	$\delta(t - t_0)$	$e^{-j2\pi f t_0}$
11.	$e^{j2\pi f_0 t}$	$\delta(f - f_0)$
12.	$\cos 2\pi f_0 t$	$\frac{1}{2}\delta(f - f_0) + \frac{1}{2}\delta(f + f_0)$
13.	$\sin 2\pi f_0 t$	$\frac{1}{2j}\delta(f - f_0) - \frac{1}{2j}\delta(f + f_0)$
14.	$u(t)$	$(j2\pi f)^{-1} + \frac{1}{2}\delta(f)$
15.	$\operatorname{sgn} t$	$(j\pi f)^{-1}$
16.	$\frac{1}{\pi t}$	$-j \operatorname{sgn}(f)$
17.	$\hat{x}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{x(\lambda)}{t - \lambda}\, d\lambda$	$-j \operatorname{sgn}(f)\, X(f)$
18.	$\sum_{m=-\infty}^{\infty} \delta(t - mT_s)$	$f_s \sum_{m=-\infty}^{\infty} \delta(f - mf_s),\ \ f_s = T_s^{-1}$

Useful Relations and Formulas

$$\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$\tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

$$\tan\left(\frac{\pi}{2} - x\right) = \cot x$$

$$\tan 2x = \frac{2\tan x}{1-\tan^2 x} = \frac{2\cot x}{\cot^2 x - 1} = \frac{2}{\cot x - \tan x}$$

$$\tan\frac{x}{2} = \pm\sqrt{\frac{1-\cos x}{1+\cos x}} = \frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$$

$$\sin x \cos y = \tfrac{1}{2}[\sin(x+y) + \sin(x-y)]$$

$$\cos x \sin y = \tfrac{1}{2}[\sin(x+y) - \sin(x-y)]$$

$$\sin x \sin y = \tfrac{1}{2}[\cos(x-y) - \cos(x+y)]$$

$$\cos x \cos y = \tfrac{1}{2}[\cos(x+y) + \cos(x-y)]$$

$$\sin x + \sin y = 2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\cos x + \cos y = 2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x - \cos y = -2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Entropy. For a random variable X defined over alphabet $\mathscr{X}$, its entropy is defined as $H(X) = -\sum_{x\in\mathscr{X}} p(x)\log_2 p(x)$.

Gaussian density. Let $X \sim \mathscr{N}(\mu, \sigma^2)$. Its pdf is $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$.

Q-function: $Q(x) = \frac{1}{\sqrt{2\pi}}\int_x^\infty e^{-z^2/2}\,dz$. For large x: $Q(x) \approx \frac{1}{\sqrt{2\pi}x}e^{-x^2/2}$.

Hilbert Transform of a signal $g(t)$: $\hat{g}(t) = \frac{1}{\pi}\int_{-\infty}^{\infty}\frac{g(\tau)}{t-\tau}\,d\tau$.

Fourier Transform Theorems

Name of Theorem		
1. Superposition (a_1 and a_2 arbitrary constants)	$a_1x_1(t)+a_2x_2(t)$	$a_1X_1(f)+a_2X_2(f)$
2. Time delay	$x(t-t_0)$	$X(f)\,e^{-j2\pi ft_0}$
3a. Scale change	$x(at)$	$\lvert a\rvert^{-1}X\left(\frac{f}{a}\right)$
b. Time reversal	$x(-t)$	$X(-f)$
4. Duality	$X(t)$	$x(-f)$
5a. Frequency translation	$x(t)\,e^{j2\pi f_0t}$	$X(f-f_0)$
b. Modulation	$x(t)\cos 2\pi f_0t$	$\frac{1}{2}X(f-f_0)+\frac{1}{2}X(f+f_0)$
6. Differentiation	$\frac{d^nx(t)}{dt^n}$	$(j2\pi f)^nX(f)$
7. Integration	$\int_{-\infty}^{t}x(t')\,dt'$	$(j2\pi f)^{-1}X(f)+\frac{1}{2}X(0)\delta(f)$
8. Convolution	$\int_{-\infty}^{\infty}x_1(t-t')\,x_2(t')\,dt'$	$X_1(f)\,X_2(f)$
9. Multiplication	$x_1(t)\,x_2(t)$	$\int_{-\infty}^{\infty}X_1(f-f')\,X_2(f')\,df'$

Differentiation Rule of Leibniz. Let $F(z)=\int_{a(z)}^{b(z)}f(x,z)\,dx$. We have

$$\frac{dF(z)}{dz}=\frac{db(z)}{dz}f(b(z),z)-\frac{da(z)}{dz}f(a(z),z)+\int_{a(z)}^{b(z)}\frac{\partial f(x,z)}{\partial z}\,dx$$

1. a) State whether each of the following statements is true or false, and justify your answer:

i) The autocorrelation function of a real wide-sense stationary (WSS) process satisfies $R_X(\tau) = R_X(-\tau)$ for all τ. [3]

ii) The output of a linear time-invariant (LTI) system driven by a wide-sense stationary Gaussian process is also a wide-sense stationary Gaussian process. [3]

iii) The mutual information $I(X;Y)$ between two discrete random variables X and Y can be negative. [2]

iv) For an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $N_0/2$, doubling the bandwidth always doubles the channel capacity. [2]

b) Consider the signal $m(t) = 3\cos(2\pi f_1 t) + 4\cos(2\pi f_2 t)$, where $f_1 = 1$ kHz and $f_2 = 3$ kHz. This signal is sampled and the samples are quantized using a uniform quantizer with 2^n levels. The dynamic range of the quantizer is set to match the peak-to-peak amplitude of the signal.

i) What is the minimum required sampling rate for the signal to be perfectly recoverable from its samples? [2]

ii) What is the signal power? [2]

iii) What is the quantization noise power? Express your answer in terms of n. [3]

iv) What is the signal-to-noise ratio (SNR) at the output of the quantizer? [2]

v) What should n be to achieve an SNR of 55.8 dB? [3]

c) Consider a zero-mean white Gaussian noise process $n(t)$ with power spectral density $S_n(f) = \frac{N_0}{2}$ for all f. This process is passed through an ideal lowpass filter with bandwidth B Hz and unit passband gain.

i) Write down the power spectral density of the output process $Y(t)$. [2]

ii) Find the autocorrelation function $R_Y(\tau)$ of the output process. [3]

iii) Find the total output noise power. [2]

d) We consider double-sideband suppressed-carrier (DSB-SC) modulation of a baseband signal $m(t)$. The signal is modulated with a carrier $\cos(2\pi f_c t)$ at frequency $f_c = 100$ Hz. The modulated waveform is corrupted by additive zero-mean Gaussian noise $n(t)$ with power spectral density

$$S_n(f) = \begin{cases} 3, & |f| \leq 150 \\ 1, & 150 < |f| \leq 300 \\ 0, & \text{otherwise.} \end{cases}$$

The message $m(t)$ is a zero-mean wide sense stationary process with power spectral density given by

$$S_m(f) = \begin{cases} 1 - \frac{|f|}{W}, & |f| \leq W \\ 0, & \text{otherwise.} \end{cases}$$

The receiver applies a bandpass filter centred at frequency f_c with bandwidth $2W$, followed by multiplication with $2\cos(2\pi f_c t)$, and then a lowpass filter of bandwidth W. Let $y(t)$ denote the output of the lowpass filter.

i) What is the total power of the noise $n(t)$? [2]

ii) Find the signal-to-noise ratio (SNR) of $y(t)$ when $W = 10$. [3]

iii) Find the SNR of $y(t)$ when $W = 80$. [3]

iv) Now suppose the receiver's local oscillator has a phase offset of $\pi/4$; that is, the receiver multiplies by $2\cos(2\pi f_c t + \pi/4)$ instead. Find the SNR of $y(t)$ when $W = 10$. [3]

2. a) Consider a zero-mean wide sense stationary (WSS) Gaussian process $X(t)$ with autocorrelation function

$$R_X(\tau) = 4e^{-2|\tau|}.$$

Define the random process

$$Z(t) = X(t)\cos(2\pi f_0 t + \Theta),$$

where Θ is a random variable uniformly distributed on $[0, 2\pi)$, independent of $X(t)$, and $f_0 > 0$ is a fixed frequency.

i) Find the mean of $Z(t)$. [3]

ii) Find the autocorrelation function $R_Z(t_1, t_2)$, and determine whether $Z(t)$ is wide sense stationary. [6]

iii) Find the power spectral density $S_Z(f)$ of $Z(t)$. [3]

iv) Find the total power of $Z(t)$. [3]

b) Consider a binary communication system that transmits a bit $b \in \{0, 1\}$ using the signals

$$s_0(t) = -A\,q(t), \quad s_1(t) = A\,q(t), \qquad 0 \le t \le T,$$

where $A > 0$ and $q(t)$ is a rectangular pulse of unit amplitude over $[0, T]$. The signal is transmitted through an AWGN channel:

$$r(t) = s_b(t) + n(t),$$

where $n(t)$ is zero-mean white Gaussian noise with power spectral density $N_0/2$. The receiver uses a matched filter followed by threshold detection with threshold λ: decide "1" if the matched filter output at time T exceeds λ, and "0" otherwise.

i) Find the distribution of the matched filter output at time T for each transmitted signal $s_b(t)$, for $b = 0, 1$. [3]

ii) Derive the bit error probability P_e as a function of the threshold λ, and express your answer in terms of the Q-function. [4]

iii) For equally likely bits, find the optimal threshold λ^* that minimizes P_e, and write down the resulting error probability. [4]

iv) Now suppose that the channel introduces a random attenuation: the received signal is $r(t) = \alpha\, s_b(t) + n(t)$, where α is a random variable independent of $n(t)$ and b.

If α takes the value 1 with probability $3/4$ and the value $1/2$ with probability $1/4$, derive the average bit error probability $\bar{P}_e$

under the optimal threshold $\lambda^* = 0$. Express your answer in terms of the Q-function. Briefly discuss the effect of fading on error performance. [4]

3. a) Assume that a memoryless source S outputs 4 symbols $\{a,b,c,d\}$ with probabilities $P(S=a)=0.4$, $P(S=b)=0.3$, $P(S=c)=0.2$, and $P(S=d)=0.1$.

i) Design a prefix-free code for source S with the minimum possible average codeword length. What is the average length of your code? Is there a unique such code? [4]

ii) What is the entropy of this source? (Use the approximations $\log_2 3 = 1.58$ and $\log_2 5 = 2.32$.) [3]

iii) The *coding efficiency* of a source code is defined as the ratio of the source entropy $H(S)$ to the average codeword length $\bar{L}$:

$$\eta = \frac{H(S)}{\bar{L}}.$$

What is the coding efficiency of the code you designed? How can you improve the coding efficiency (i.e., bring the average code length closer to the entropy)? [3]

iv) Suppose the same code is used for a different source T with probabilities $P(T=a)=0.1$, $P(T=b)=0.2$, $P(T=c)=0.3$, $P(T=d)=0.4$. What is the average codeword length for source T? Is this code still optimal for T? [4]

b) *Binary Erasure Channel (BEC).* Consider a memoryless channel with binary input $X \in \{0,1\}$ and ternary output $Y \in \{0,e,1\}$, with transition probabilities:

$$P(Y=x\,|\,X=x)=1-\varepsilon, \qquad P(Y=e\,|\,X=x)=\varepsilon, \qquad \text{for } x\in\{0,1\},$$

where $\varepsilon \in (0,1)$ is the *erasure probability*. That is, each transmitted bit is either received correctly (with probability $1-\varepsilon$) or replaced by an erasure symbol e (with probability ε). Let the input distribution be $P(X=0)=p$ and $P(X=1)=1-p$.

i) Compute the conditional entropy $H(Y|X)$ as a function of ε. [3]

ii) Compute the entropy $H(Y)$ as a function of p and ε. [4]

iii) Find the channel capacity C of the BEC by maximizing the mutual information $I(X;Y)=H(Y)-H(Y|X)$ over the input distribution p. [5]

iv) Compare the capacity of the BEC with erasure probability ε to the capacity of a binary symmetric channel (BSC) with the same parameter ε as the crossover probability. Which channel has higher capacity for $0 \le \varepsilon \le 1/2$? Give an intuitive explanation for your answer. [4]